Project Charter Plan

Intelligent System for Pakistan Sign Language Learning and Translation

FAST NUCES - Karachi Campus
Shah Latif Town
Karachi, Sindh

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Table of Contents

Executive Summary

The proposed project addresses the communication challenges faced by the deaf and hard-of-hearing community in Pakistan due to the limited availability of digital tools supporting Pakistan Sign Language (PSL). While sign language translation technologies have advanced globally, most systems focus on American Sign Language (ASL) or British Sign Language (BSL), leaving PSL underrepresented. This project aims to design and develop a mobile application that provides real-time PSL translation, an integrated dictionary module, and structured learning and assessment features. By leveraging an iterative development model with encoder–decoder deep learning architecture, computer vision frameworks, and a scalable technology stack, the system will deliver an accessible and user-friendly solution. Ultimately, this application will bridge communication gaps, enhance learning opportunities, and contribute to the social inclusion of the deaf community in Pakistan.

Project Purpose/Justification

Business Need/Case

The PSL Translation and Learning Application project has been created to address the lack of digital tools supporting Pakistan Sign Language (PSL), which results in communication barriers between the deaf community and the hearing population. While international sign languages have seen significant advancements in translation and learning technologies, PSL remains underrepresented. This creates a strong social need to provide equal access to communication, education, and inclusion opportunities for the deaf community in Pakistan. The resources invested in the design and implementation of this application will be justified by the resulting benefits of enhanced accessibility, greater awareness of Pakistan Sign Language.

Business Objectives

The business objectives for this project are in direct support of promoting accessibility, inclusivity, and technological advancement in line with organizational and social development goals.

- Design and test a PSL translation module capable of PSL-to-text/speech and text/speech-to-PSL within the next 6 months
- Complete implementation of a structured PSL learning and assessment module within 8 months.

Project Description

The “Intelligent System for Pakistan Sign Language Learning and Translation” project will provide an accessible mobile application designed to bridge the communication gap between hearing and non-hearing individuals in Pakistan. The system will combine modern technologies in computer vision and deep learning to enable real-time translation between Pakistan Sign Language (PSL) and text or speech. In addition, the application will include a structured dictionary, interactive lessons, and assessment features to support effective PSL learning. The project is intended to promote inclusivity, increase awareness of PSL, and provide an affordable digital platform for communication and education within the community.

Project Objectives and Success Criteria

The objectives which mutually support the milestones and deliverables for this project have been identified. In order to achieve success on the project, the following objectives must be met within the designated time:

- To develop a translation module for PSL-to-text/speech and text/speech-to-PSL.
- To create a structured learning module with lessons and practice sessions.
- To implement an assessment system with quizzes and sign replication tasks.
- To build a categorized PSL dictionary for easy reference.

Requirements

This project must meet the following list of requirements in order to achieve success.

- The application must provide real-time PSL-to-text/speech and text/speech-to-PSL translation with reliable accuracy.
- A PSL dictionary must be developed, categorized alphabetically and chapter-wise, containing a sufficient number of signs for practical use.
- The learning module must include structured lessons, tutorials, and practice exercises to support step-by-step PSL learning.
- An assessment system must be implemented to evaluate user progress through quizzes, sign replication tasks, and chapter-level tests.
- The application must be deployed on mobile platforms (Android) with an intuitive and user-friendly interface.
- The solution must be tested thoroughly with stakeholders prior to full deployment.
- Data must be securely stored and managed ensuring user privacy and reliable access.

Additional requirements may be added as necessary, with project sponsor approval, as the project moves forward.

Constraints

The following constraints pertain to the PSL Translation and Learning project:

- The project scope is limited to Pakistan Sign Language (PSL) only; support for other sign languages is out of scope for this phase.

- Resource costs will primarily involve time, dataset preparation, and model training.

Assumptions

The following are a list of assumptions. Upon agreement and signature of this document, all parties acknowledge that these assumptions are true and correct:

- It assumes the availability of sufficient PSL datasets to support accurate training of the AI models.
- It assumes that users will have access to smartphones and reliable internet connectivity.
- The effectiveness of the system depends on the cultural acceptance of the application by the Deaf community, as accuracy and representation of PSL signs must be validated.
- The project depends on potential collaborations with NGOs, schools, and government institutions for further enhancements and the project's sustainability.

Preliminary Scope Statement

The PSL Translation and Learning project will include the design, development, testing, and delivery of a mobile application that provides real-time Pakistan Sign Language (PSL) translation, an integrated dictionary, structured learning modules, and assessment features. All technical resources, including personnel, software frameworks, and database management systems, will be managed by the project team. Project activities will be carried out independently of daily academic and organizational operations, and all required testing will be conducted with pilot users and domain experts, including members of the Deaf Community. Project funding and resources will be managed by the project manager within the allocated limits defined in this document, with any additional requirements subject to sponsor approval. This project will be considered complete when the application is deployed on mobile platforms, all planned modules are functional, user testing and validation are successfully completed, technical documentation is finalized, and the final project report is submitted for approval.

Risks

There are various risks associated with the project. Firstly, the accuracy of the trained model depends on the model chosen and its current capability. Such a risk can be mitigated through adopting an iterative approach in model training and development. Secondly, validation from the Deaf community and user acceptance upon release is also a risk that can be addressed through iterative development. Furthermore, the availability of Dataset for PSL is limited as there is a lack of such resources, especially from a technological point of view. Finally, the lack of availability of smartphones, internet connectivity, and other such access issues for the intended users is also a potential risk.

Project Deliverables

The following deliverables must be met upon the successful completion of the project. Any changes to these deliverables must be approved by the project sponsor.

- The application must provide real-time PSL-to-text/speech and text/speech-to-PSL translation with reliable accuracy.
- A PSL dictionary must be developed, categorized alphabetically and chapter-wise, containing a sufficient number of signs for practical use.
- The learning module must include structured lessons, tutorials, and practice exercises to support step-by-step PSL learning.
- An assessment system must be implemented to evaluate user progress through quizzes, sign replication tasks, and chapter-level tests.
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Summary Milestone Schedule

The project Summary Milestone Schedule is presented below. As requirements are more clearly defined this schedule may be modified. Any changes will be communicated through project status meetings by the project manager.

Summary Milestone Schedule – List key project milestones relative to project start.	
Project Milestone	**Target Date (mm/dd/yyyy)**
• Project Start	09/10/2025
• SRS	10/31/2025
• SDS	12/15/2025
• Authentication and Authorization Module	12/15/2026
• Dictionary Module	12/31/2025
• Translation Module	01/31/2026
• Learning Module	03/10/2026
• Assessment Module	03/10/2026
• Complete Solution Simulation and Testing	04/30/2026
• Deploy Solution	05/30/2026

• Project Complete	06/01/2026

Summary Budget

The following table contains a summary budget based on the planned cost components and estimated costs required for successful completion of the project.

Summary Budget – List component project costs	
Project Component	**Component Cost**
• **External Consultation**	$1013
• **Data set (collection + augmentation)**	$2025
• **Hardware (GPU etc.)**	$2700
• **Cloud Storage**	$1350
• **Software Management Tools**	$270
• **Software Licences / Development Tools**	$608
• **Employee Training**	$878
• **Office Supplies**	$203
• **Office Operational Costs (Transport, Electricity etc.)**	$203
• **Research and Development (AI model iterations)**	$1013
• **Product Operational Cost (testing, deployment)**	$1688
• **System Updates & Maintenance**	$2025
• **External Consultation**	$2160
Total	**$17933**

Project Approval Requirements

Success for the PSL Translation and Learning project will be achieved when a fully functional and tested mobile application is developed and deployed, meeting the requirements outlined in this charter. This includes the delivery of the translation module, structured learning module, assessment system, and PSL dictionary within the allocated time.

Additionally, success will be measured by ensuring cultural accuracy, accessibility for diverse users, and validation by the Deaf community. The project will be deemed complete and successful once the final application, supporting documentation, and demonstration of key functionalities are delivered to the stakeholders.

Approval and sign-off on project completion will be determined by the Project Sponsor, who will authorize the closeout of the project once all deliverables and success criteria have been met.

Project Manager

Alizah Basit is named Project Manager for the duration of the "Intelligent System for Pakistan Sign Language Learning and Translation" Project. Alizah Basit's responsibility is to oversee all project activities, including task management, scheduling, communication with stakeholders, and ensuring the project remains within scope, budget, and timeline.

The project team will include members specializing in artificial intelligence, computer vision, natural language processing, and mobile application development. Alizah Basit will coordinate all resource requirements, including datasets, computing infrastructure, and development tools, through the academic supervisor and supporting faculty.

AUTHORIZATION

Approved by the Project Sponsor:

Date: September 20, 2025

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